1. Introduction

The purpose of this phase is to formally specify the architecture and functionality of the Smart Cultural Heritage Guide application. Building on the concept definition, user research and low-fidelity prototype developed in previous phases, this document defines the system requirements, architecture, modules and UML specifications.

The application aims to enhance cultural tourism experiences by providing interactive maps, multimedia content, walking tours and context-aware recommendations based on the user's location and preferences.

2. System Overview

The **Smart Cultural Heritage Guide** is a mobile application that helps users discover and explore cultural heritage sites such as museums, monuments, historical landmarks and archaeological locations.

The system provides:

- Location-based recommendations;
- Interactive maps;
- Walking tours;
- Multimedia content (text, images, audio and video);
- Personalized experiences;
- Context-aware adaptation using GPS and environmental data.

3. Refined Functional Requirements

ID	Description
FR1	The system shall detect the user's location using GPS.
FR2	The system shall display nearby cultural attractions on an interactive map.
FR3	The system shall provide detailed information about cultural locations.
FR4	The system shall support multimedia content including text, images, audio and video.
FR5	The system shall allow users to search for cultural sites.
FR6	The system shall generate and display walking tours.
FR7	The system shall provide route navigation.
FR8	The system shall allow users to save favourite locations.
FR9	The system shall provide personalized recommendations.

ID	Description
FR10	The system shall adapt content presentation according to contextual information.

Table 1: Functional Requirements

4. Non-Functional Requirements

Category	Requirement
Usability	Easy and intuitive interface
	Minimal learning curve
	Accessible navigation
Performance	Fast loading times
	Real-time GPS updates
	Smooth multimedia playback
Reliability	Stable operation during visits
	Accurate location services
Security	Secure storage of user data
	Privacy protection for location information
Compatibility	Android support
	iOS support
	Responsive layouts
Accessibility	Adjustable text size
	Audio alternatives
	High-contrast interface

Table 2: Non-Functional Requirements

5. Use Cases

Use Case 1: Explore Nearby Cultural Places

- Actor: User
- Preconditions: GPS is enabled.

- Main Flow:
  - User opens the application;
  - System retrieves location;
  - Nearby attractions are displayed;
  - User selects an attraction;
  - System displays detailed information.
- Postcondition: User receives information about the selected attraction.

Use Case 2: Start Walking Tour

- Actor: User
- Preconditions: A tour is available.
- Main Flow:
  - User selects a tour;
  - System generates the route;
  - Navigation begins;
  - Information is displayed at each point of interest.
- Postcondition: Tour is completed successfully.

Use Case 3: Explore Museum

- Actor: User
- Preconditions: User enters museum mode.
- Main Flow:
  - User selects an exhibit;
  - System retrieves exhibit information;
  - Multimedia content is displayed;
  - User saves favourites if desired.
- Postcondition: Exhibit information is explored.

Use Case 4: Manage Preferences

- Actor: User
- Main Flow:
  - User opens settings;
  - User selects interests and preferences;
  - System updates recommendations.
- Postcondition: Personalized content is configured.

6. UML Use Case Diagram

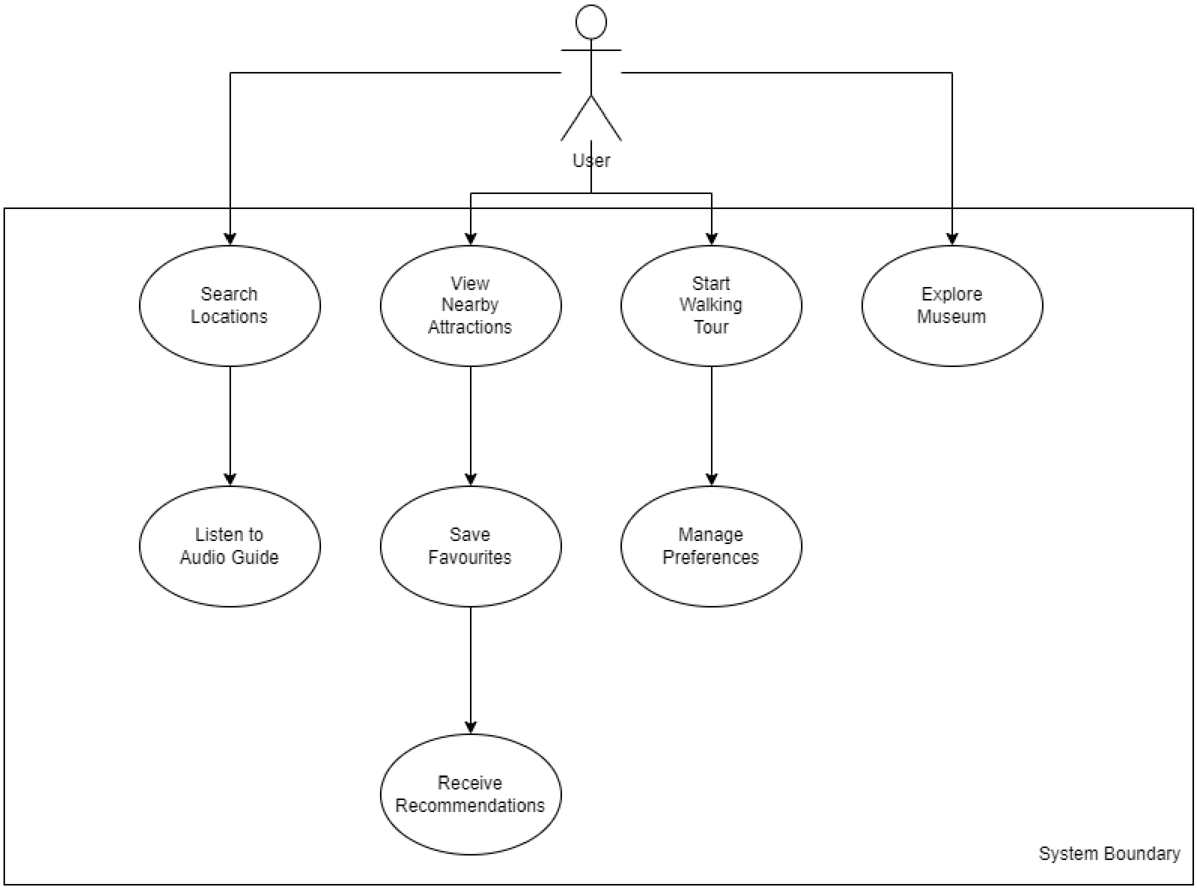

Figure 1: Use Case Diagram

7. UML Component Diagram

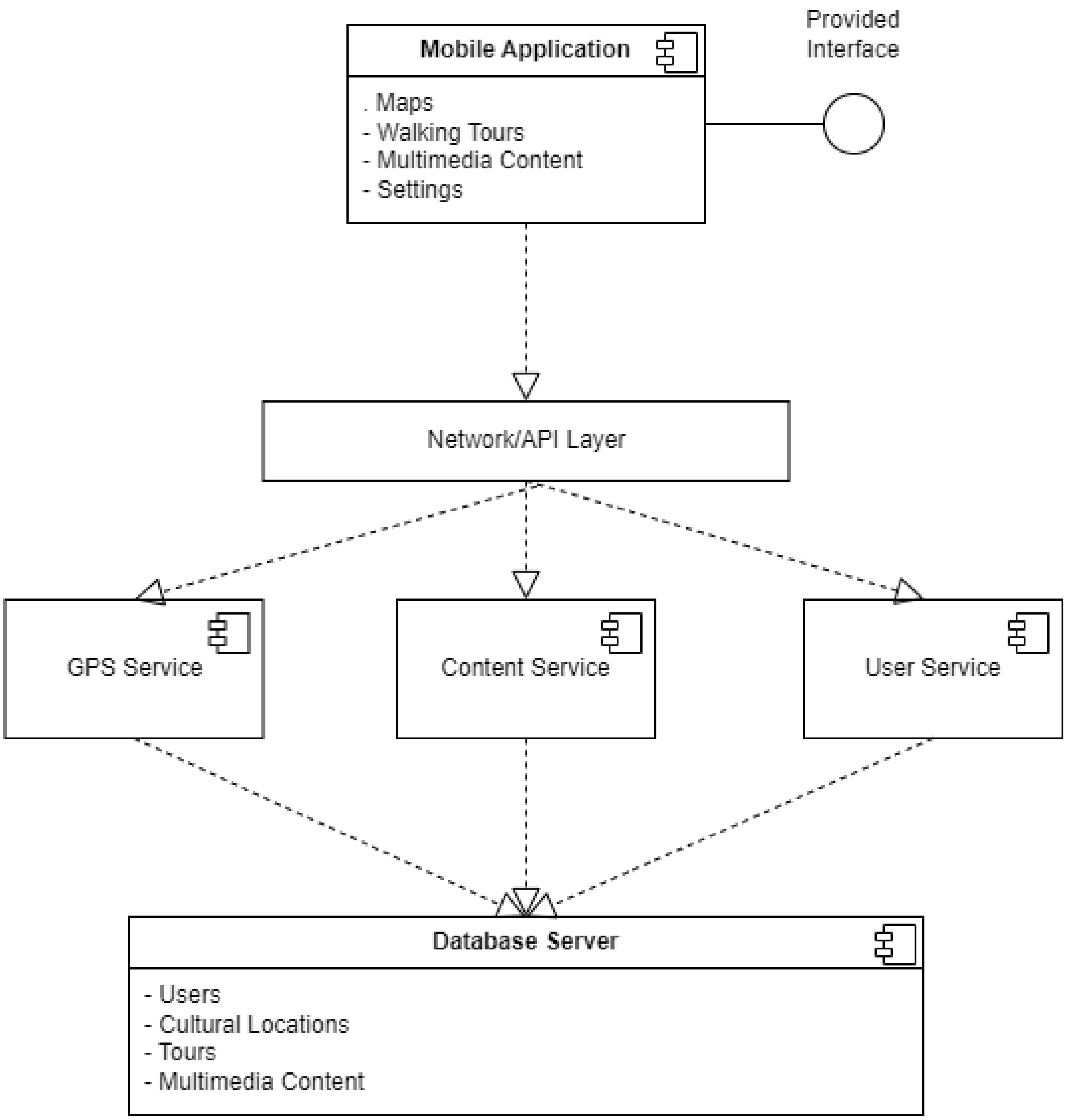

Figure 2: Component Diagram

8. UML Sequence Diagram

Explore Nearby Cultural Places

```
User              Mobile App           GPS Service            Backend           Database
  |                    |                    |                    |                  |
  | Open App           |                    |                    |                  |
  |------------------->|                    |                    |                  |
  |                    | Get Location       |                    |                  |
  |                    |------------------->|                    |                  |
  |                    |<-------------------|                    |                  |
  |                    | Request POIs       |------------------->|                  |
  |                    |                    |                    | Query            |
  |                    |                    |                    |----------------->|
  |                    |                    |                    |<-----------------|
  |                    |<-------------------|                    |                  |
  | View Results       |                    |                    |                  |
  |<-------------------|                    |                    |                  |
```

Figure 3: Sequence Diagram for exploring nearby cultural places

Start Walking Tour

```
User              Mobile App            Backend             Database
  |                    |                    |                    |
  | Select Tour        |                    |                    |
  |------------------->|                    |                    |
  |                    | Request Route      |                    |
  |                    |------------------->|                    |
  |                    |                    | Get Route          |
  |                    |                    |------------------->|
  |                    |                    |<-------------------|
  |                    |<-------------------|                    |
  | Start Tour         |                    |                    |
  |<-------------------|                    |                    |
```

Figure 4: Sequence Diagram for starting a walking tour

9. UML Class Diagram

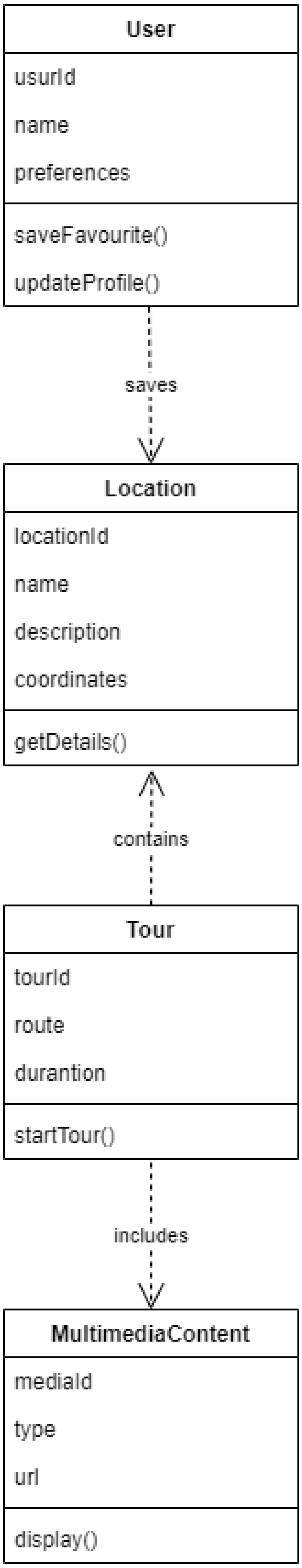

Figure 5: Class Diagram

10. System Modules

The Smart Cultural Heritage Guide is organized into several functional modules, each responsible for a specific set of tasks. This modular architecture improves maintainability, scalability and system performance while ensuring a clear separation of responsibilities.

10.1 User Interface Module

The User Interface (UI) Module is responsible for all interactions between the user and the application. It provides an intuitive and accessible interface through which users can access the system's functionalities.

Responsibilities

- Display interactive maps and cultural locations;
- Present multimedia content (text, images, audio, and video);
- Provide navigation between application screens;
- Manage user settings and preferences;
- Display personalized recommendations and walking tours.

Inputs

- User actions (touch gestures, button selections, searches);
- Data received from backend services.

Outputs

- Visual and multimedia information presented to the user.

10.2 Sensor Data Collection Module

The Sensor Data Collection Module gathers contextual information from the device sensors to support location-based and adaptive functionalities.

Responsibilities

- Collect GPS location data;
- Monitor environmental information when available;
- Provide real-time contextual information to the system.

Inputs

- GPS sensor data;
- Device environmental sensors.

Outputs

- User location information;
- Contextual data used for recommendations and adaptations.

10.3 Data Processing Module

The Data Processing Module is responsible for analysing collected data and generating personalized results for users.

Responsibilities

- Process location and user preference data;
- Generate personalized recommendations;
- Select relevant cultural content;
- Support context-aware adaptation of information.

Inputs

- Sensor data;
- User preferences;
- Cultural content data.

Outputs

- Recommended locations and tours;
- Personalized content and adaptive responses.

10.4 Multimedia Content Module

The Multimedia Content Module manages all cultural information and media resources presented within the application.

Responsibilities

- Store and organize cultural content;
- Manage text descriptions, images, audio guides, and videos;
- Deliver multimedia resources to users.

Inputs

- Content requests from the application;
- Multimedia data stored in the backend.

Outputs

- Multimedia content displayed to users.

10.5 Network Communication Module

The Network Communication Module enables communication between the mobile application and backend services.

Responsibilities

- Send requests to backend services;
- Retrieve cultural information and user data;
- Synchronize application data;
- Manage communication between client and server.

Inputs

- Requests from the mobile application.

Outputs

- Responses received from backend services.

10.6 Backend Server Module

The Backend Server Module manages the application's business logic and data storage services.

Responsibilities

- Manage user accounts and preferences;
- Store cultural heritage information;
- Manage walking tours and routes;
- Handle content requests from the mobile application;
- Coordinate communication with the database.

Inputs

- Requests from the Network Communication Module.

Outputs

- Cultural information;
- User profile data;
- Tour and recommendation data.

10.7 Database Module

The Database Module stores and manages all persistent information required by the system.

Responsibilities

- Store user information and preferences;
- Store cultural heritage locations;
- Store multimedia content;
- Store route and tour information;
- Support efficient data retrieval.

Inputs

- Data received from backend services.

Outputs

- Requested information returned to the backend.